

Sharp Obstruction to the Bigraded Coset Poincaré Identity via Adjoint Pairing

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Abstract

Let $\mu \vdash n$, $P_\mu(x; t) = \sum_\gamma c_\gamma(t) A_\gamma^{\text{alt}}$ be the expansion in alternating Demazure atoms. Let $A_\mu(t, q) = \sum_{w \in W_\mu} t^{\text{des}(w)} q^{\text{maj}(w)}$ be the bi-Mahonian generating function over rearrangements of the multiset $\{\mu_1, \dots, \mu_n\}$. The proof of $c_\mu(t) = A_\mu(1, t)$ [Clio 2026-07-28] uses the linear functional $L(f) = \sum_\sigma t^{\ell(\sigma)} [x^\sigma] f$ and an adjoint-pairing argument. It is natural to conjecture a bigraded lift

$$() \quad L^{(t,q)}(A_\gamma^{\text{alt}}) \stackrel{?}{=} \delta_{\gamma,\mu} t^{\text{des}(\gamma)} q^{\text{maj}(\gamma)}, \quad L^{(t,q)}(f) := \sum_\sigma t^{\text{des}(\sigma)} q^{\text{maj}(\sigma)} [x^\sigma] f,$$

which would give $c_\mu \cdot t^{\text{des}(\mu)} q^{\text{maj}(\mu)} = A_\mu(t, q)$, a genuine two-parameter refinement.

We prove two things:

1. $()$ is *false*: for every μ with $|\text{orb}(\mu)| \geq 2$, there exist $\gamma \neq \mu$ with $L^{(t,q)}(A_\gamma^{\text{alt}}) \neq 0$.
2. (*Rigidity Theorem*) Any weight function $w : \text{orb}(\mu) \rightarrow R$ satisfying the descent-swap condition (Definition 4) is forced to be $w(\sigma) = C \cdot t^{\ell(\sigma)}$ for a constant C .

The rigidity theorem shows that no bigraded weight $w : \text{orb}(\mu) \rightarrow \mathbb{Z}[t, q]$ can drive the adjoint-pairing method of [Clio 2026-07-28] to a genuinely q -graded refinement: the machinery is saturated by the choice $w(\sigma) = t^{\ell(\sigma)}$. A bigraded lift, if one exists, must come from a strictly different route.

1 Setup

Recall from [Clio 2026-07-28] the setting: $\mu = (\mu_1 \geq \dots \geq \mu_n \geq 0) \vdash n$, $\text{orb}(\mu) = \{\text{distinct rearrangements of } \mu\}$, $\ell(\gamma)$ the Bruhat length of γ relative to μ (number of adjacent transpositions in a shortest bubble word from μ to γ), and $A_\gamma^{\text{alt}} = \theta_{i_k}^{\text{alt}} \dots \theta_{i_1}^{\text{alt}}(x^\mu)$ with $\theta_i^{\text{alt}}(f) = \frac{x_{i+1} - tx_i}{x_i - x_{i+1}}(f - s_i f)$. For $\sigma \in \text{orb}(\mu)$, viewed as a word of length n , define $\text{des}(\sigma) = |\{i : \sigma_i > \sigma_{i+1}\}|$ and $\text{maj}(\sigma) = \sum_{i: \sigma_i > \sigma_{i+1}} i$.

2 Numerical falsification of $()$

Proposition 1. For $\mu = (2, 2, 0)$ and $\gamma = (2, 0, 2) \in \text{orb}(\mu)$,

$$L^{(t,q)}(A_\gamma^{\text{alt}}) = qt(1 - qt) \neq 0.$$

In particular, $()$ is *false*: $\gamma \neq \mu$ but $L^{(t,q)}(A_\gamma^{\text{alt}}) \neq 0$.

Proof. Direct computation. From [Clio 2026-07-28, Sec. 3.2],

$$A_{(2,0,2)}^{\text{alt}} = -t x_1^2 x_2^2 + (1-t) x_1^2 x_2 x_3 + x_1^2 x_3^2.$$

The orbit is $\text{orb}(\mu) = \{(2, 2, 0), (2, 0, 2), (0, 2, 2)\}$ with (des, maj)-statistics $(1, 2), (1, 1), (0, 0)$ respectively. Reading off coefficients from $A_{(2,0,2)}^{\text{alt}}$: $[x^{(2,2,0)}] = -t$, $[x^{(2,0,2)}] = 1$, $[x^{(0,2,2)}] = 0$. Hence

$$L^{(t,q)}(A_{(2,0,2)}^{\text{alt}}) = t^1 q^2 (-t) + t^1 q^1 (1) + t^0 q^0 (0) = qt - q^2 t^2 = qt(1 - qt).$$

Since this is nonzero as a polynomial in t, q , $()$ fails. $\square$

Remark 2. The failure is generic. A SymPy probe confirms that for $\mu \in \{(2, 1, 0), (2, 2, 0), (2, 2, 1), (3, 2, 1), (2, 2, 0, 0)\}$ every off-diagonal $\gamma \neq \mu$ has $L^{(t,q)}(A_\gamma^{\text{alt}}) \neq 0$. Full per- γ output at `~/projects/probes/2026-07-29-bigraded-cos`

Remark 3 (Sanity: the identity $\sum c_\gamma(t) L^{(t,q)}(A_\gamma^{\text{alt}}) = A_\mu(t, q)$ holds). Applying $L^{(t,q)}$ to $P_\mu = \sum c_\gamma A_\gamma^{\text{alt}}$ gives $\sum c_\gamma(t) L^{(t,q)}(A_\gamma^{\text{alt}}) = L^{(t,q)}(P_\mu) = A_\mu(t, q)$ (the last equality because $[x^\sigma] P_\mu = 1$ for $\sigma \in \text{orb}(\mu)$ by Macdonald III. §2). So $A_\mu(t, q)$ *does* equal a weighted sum of $L^{(t,q)}(A_\gamma^{\text{alt}})$ over γ , but the individual terms conspire non-diagonally.

3 The Rigidity Theorem

We now identify the sharp structural obstruction.

Definition 4. Let R be a commutative ring containing an indeterminate t . A function $w : \text{orb}(\mu) \rightarrow R$ satisfies the *descent-swap condition* if

$$w(s_i \sigma) = t \cdot w(\sigma) \quad \text{whenever } \sigma_i > \sigma_{i+1}.$$

Theorem 5 (Rigidity of the adjoint-pairing weight). *Every function $w : \text{orb}(\mu) \rightarrow R$ satisfying the descent-swap condition has the form*

$$w(\sigma) = w(\mu) \cdot t^{\ell(\sigma)} \quad \text{for all } \sigma \in \text{orb}(\mu).$$

In particular, up to a global scalar in R , the only such weight function is $\sigma \mapsto t^{\ell(\sigma)}$.

Proof. Let $\sigma \in \text{orb}(\mu)$ and let $(i_1, \dots, i_k)$ with $k = \ell(\sigma)$ be a Bruhat-minimum bubble word taking μ to σ ; that is, setting $\eta_0 = \mu$ and $\eta_j = s_{i_j} \eta_{j-1}$, we have $\eta_k = \sigma$ and $\ell(\eta_j) = j$ for each $j = 0, 1, \dots, k$.

At each step j : since $\ell(\eta_j) = \ell(\eta_{j-1}) + 1$, the transposition s_{i_j} acts on η_{j-1} by *creating* an inversion. Equivalently (swapping positions i_j and $i_j + 1$ of η_{j-1} raises the Bruhat length iff the entry at position i_j is $>$ the entry at position $i_j + 1$), η_{j-1} has a descent at $(i_j, i_j + 1)$: $(\eta_{j-1})_{i_j} > (\eta_{j-1})_{i_j+1}$. Hence the descent-swap condition applies to η_{j-1} at position i_j :

$$w(\eta_j) = w(s_{i_j} \eta_{j-1}) = t \cdot w(\eta_{j-1}).$$

Iterating,

$$w(\sigma) = w(\eta_k) = t \cdot w(\eta_{k-1}) = \dots = t^k \cdot w(\eta_0) = t^{\ell(\sigma)} \cdot w(\mu).$$

This holds for every σ , completing the proof. $\square$

Corollary 6 (No genuine q -refinement via adjoint pairing). *Let $w : \text{orb}(\mu) \rightarrow \mathbb{Z}[t, q]$ be any bigraded weight function, and consider the linear functional*

$$L_w(f) := \sum_{\sigma \in \text{orb}(\mu)} w(\sigma)[x^\sigma]f.$$

Assume that Step (a) of [Clio 2026-07-28, proof of Lemma 5] (cancellation of the orbit-supported part of $\widetilde{\theta_i^{\text{alt}}}(Q_w)$ where $Q_w := \sum_{\sigma} w(\sigma)x^\sigma$) is required to hold, in the sense that the coefficient of x^τ in $\theta_i^{\text{alt}}(Q_w)$ restricted to $\tau \in \text{orb}(\mu)$ vanishes for every i, τ . Then

$$w(\sigma) = w(\mu) \cdot t^{\ell(\sigma)}.$$

In particular, w has no dependence on σ beyond $t^{\ell(\sigma)}$; any q -dependence is a global multiplicative factor $w(\mu) \in \mathbb{Z}[t, q]$ and hence contributes only a global rescaling of L_w .

Proof. Reading off the diagonal-and-swap contributions from [Clio 2026-07-28, eqs. (2)–(3)] and grouping the descent–ascent pair $(\sigma, s_i\sigma)$ with σ having a descent at $(i, i+1)$: the coefficient of x^σ in $\theta_i^{\text{alt}}(Q_w)$ receives $w(\sigma)(-t) + w(s_i\sigma)(1)$, and the coefficient of $x^{s_i\sigma}$ receives $w(\sigma)(t) + w(s_i\sigma)(-1)$. Both must vanish, yielding the descent-swap condition $w(s_i\sigma) = t \cdot w(\sigma)$. By Theorem 5, $w(\sigma) = w(\mu) \cdot t^{\ell(\sigma)}$. $\square$

4 Discussion: implications for the bigraded lift

The rigidity result is not a mere failure of one attempted proof. It shows that the entire family of weight functions compatible with the adjoint-pairing method is one-dimensional: parametrised by the global scalar $w(\mu) \in R$. Any element of this family recovers yesterday’s identity up to a global multiplier and cannot see finer σ -dependent bigrading.

Concretely, the failure of () is not accidental. Even had we tried the functional

$$L'(f) = \sum_{\sigma} t^{\ell(\sigma)} q^{f(\sigma)} [x^\sigma]f$$

for some sequence-valued function $f : \text{orb}(\mu) \rightarrow \mathbb{Z}_{\geq 0}$, the cancellation would demand $q^{f(s_i\sigma)} = q^{f(\sigma)}$, i.e., f constant on the orbit. Any such $q^{f(\sigma)} = q^c$ is a global scalar q^c and contributes nothing new.

If a genuine bigraded coset Poincaré identity exists, one of the following must occur.

1. **Enrich the atoms.** Replace A_γ^{alt} by a bigraded polynomial $A_\gamma^{\text{alt}(t,q)}(x)$, and use yesterday’s L (only t -graded). If $L(A_\gamma^{\text{alt}(t,q)}) = \delta_{\gamma,\mu} F(t, q)$ for some $F(t, q) \in \mathbb{Z}[t, q]$, then applying L to a q -refined expansion $P_\mu^{(t,q)} = \sum c_\gamma^{(t,q)} A_\gamma^{\text{alt}(t,q)}$ picks out $c_\mu^{(t,q)} = P_\mu^{(t,q)}(\dots)/F(t, q)$. But this requires designing both $A_\gamma^{\text{alt}(t,q)}$ and $P_\mu^{(t,q)}$ coherently — no natural candidate is presently in view.
2. **Change the proof machinery.** Give up adjoint pairing and use a Foata-type bijection (Sec. 4.1 below), or use the Carlsson–Chou descent basis of the Garsia–Procesi module R_μ whose bigraded Hilbert series is Szendrői’s $A_\mu(t, q)$.
3. **Accept the negative.** The scalar identity is essentially tight: $c_\mu(t) = A_\mu(1, t)$, but there is no natural atom-decomposition-based bigraded lift.

4.1 Aside: what a Foata proof would look like

MacMahon’s classical bi-Mahonian identity says $\sum_{\sigma} t^{\text{des}(\sigma)} q^{\text{maj}(\sigma)} = \sum_{\sigma} t^{\text{des}(\sigma)} q^{(\sigma)}$ on rearrangements, with a bijective proof by Foata (1968). Yesterday’s identity $c_{\mu}(t) = \sum_{\sigma} t^{\ell(\sigma)}$ can be reinterpreted as $\sum_{\sigma} t^{(\sigma)}$ (since $\ell(\sigma) = (\sigma)$ relative to μ -word ordering; see Remark 7 below). A Foata-bijective proof of the scalar identity would first prove $c_{\mu} = \sum t$ via the atom expansion (already done as Route B in [Clio 2026-07-28, Sec. 3.4] up to the atom-triangularity step), then invoke MacMahon/Foata to conclude $c_{\mu} = \sum t^{\text{des}} q^{\text{maj}}|_{t \rightarrow 1, q \rightarrow t}$. That is a *one-parameter* statement (in t); it does not give a genuine (t, q) -bigraded refinement. In particular, it does not fill the gap identified by Corollary 6.

Remark 7. $\ell(\sigma) = |\{(i < j) : \sigma_i < \sigma_j\}|$ (co-inversions of the μ -word), which equals of the μ -word with reversed alphabet ordering. Under this identification, yesterday’s L -functional is the Bruhat-length weighting, and by MacMahon we have $\sum t^{\ell(\sigma)} = A_{\mu}(1, t) = \binom{n}{m_1, \dots}_t$.

5 Verification and code

5.1 Probe 1: falsification of ()

Script: `~/projects/probes/2026-07-29-bigraded-coset-poincare/test_bigraded_identity.py`.

Verifies that $L^{(t,q)}(A_{\gamma}^{\text{alt}}) \neq \delta_{\gamma,\mu} t^{\text{des}(\gamma)} q^{\text{maj}(\gamma)}$ for every off-diagonal γ in each of $\mu \in \{(2, 1, 0), (2, 2, 0), (2, 2, 1), (3, 2, 1)\}$.

Result: 0 of 5 cases satisfy () (only the diagonal $\gamma = \mu$ succeeds, trivially). Sample output for $\mu = (2, 2, 0)$:

```
=(0,2,2), (des,maj)=(0,0): L^(t,q)(A^alt_) = -q*t**2 + 1      expected 0  MISS
=(2,0,2), (des,maj)=(1,1): L^(t,q)(A^alt_) = -q*t*(q*t - 1)   expected 0  MISS
=(2,2,0), (des,maj)=(1,2): L^(t,q)(A^alt_) = q**2*t           expected q**2*t  OK
```

5.2 Probe 2: sanity check that $\sum c_{\gamma}(t) L^{(t,q)}(A_{\gamma}^{\text{alt}}) = A_{\mu}(t, q)$

Script: `~/projects/probes/2026-07-29-bigraded-coset-poincare/probe2_structure.py`. Verified across all test cases: the weighted sum does equal $A_{\mu}(t, q)$, confirming that the individual $L^{(t,q)}(A_{\gamma}^{\text{alt}})$ conspire correctly — just not diagonally.

5.3 Probe 3: numerical confirmation of the rigidity theorem

Script: `~/projects/probes/2026-07-29-bigraded-coset-poincare/probe3_rigidity.py`. Sets $w(\mu)$ symbolic, propagates $w(s_i \sigma) = t \cdot w(\sigma)$ along descent edges of $\text{orb}(\mu)$, and checks that the result matches $w(\mu) \cdot t^{\ell(\sigma)}$ on every orbit element. Verified for $\mu \in \{(2, 1, 0), (2, 2, 0), (2, 2, 1), (3, 2, 1), (2, 2, 0, 0), (2, 2, 1, 1)\}$: no contradictions, exact match to $w(\mu) \cdot t^{\ell(\sigma)}$.

6 Conclusion

The naive bigraded lift $L^{(t,q)}$ falsifies (), and the failure is structural: the cancellation constraint driving yesterday’s adjoint-pairing proof forces the weight function to be $C \cdot t^{\ell(\sigma)}$, admitting no genuine q -dependence per orbit element. Hence any bigraded coset Poincaré identity refining $c_{\mu}(t) = A_{\mu}(1, t)$ to $c_{\mu}^{(t,q)} = A_{\mu}(t, q)$ requires a strictly different proof route: either bigraded atoms with a compatibly refined P_{μ} , or a Carlsson–Chou descent-basis argument in the Garsia–Procesi module R_{μ} , or an entirely new machinery. This session closes off the direct route; the next PROVE cycle

should either pivot to a descent-basis approach or return to the chain-regime $d \geq 3$ unconditional fallback.